Engineering Mechanics: ME101

Lecture 7 : Distributed Forces and Centre of Gravity

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Distributed forces: Center of Gravity

- The earth exerts a gravitational force on each of the particles forming a body. These forces can be replace by a single equivalent force equal to the weight of the body and applied at the *center of gravity* for the body.

- The *centroid of an area* is analogous to the center of gravity of a body. The concept of the *first moment of an area* is used to locate the centroid.

- Determination of the area of a *surface of revolution* and the volume of a *body of revolution* are accomplished with the *Theorems of Pappus-Guldinus*.

Introduction

- ❖ The **center of mass** is the **mean position** of the mass **in an** object.

- ✓ Centre of mass is the **point at which the distribution of mass is equal in all directions**, and does not depend on gravitational field.

- ❖ Then there's the **center of gravity**, which is the point where gravity appears to act. For many objects, these two points are in exactly the same place. But they're only the same when the gravitational field is uniform across an object.

- ✓ Centre of gravity is the point at which the **distribution of weight is equal in all directions**, and does depend on gravitational field.

- ❖ **Centroid** is the **geometric centre of a body**.

- ✓ The **Center of Gravity** is the **same** as the **centroid** when the density is the **same** throughout.

- ✓ The centroid of a solid is similar to the center of mass. However, calculating the centroid involves only the geometrical shape of the solid. The center of gravity will equal the centroid if **the body is homogenous i.e. constant density.**

Centre of Gravity

∗ A body of mass m in equilibrium under the action of tension in the cord, and resultant W of the gravitational forces acting on all particles of the body.
- The resultant is collinear with the cord

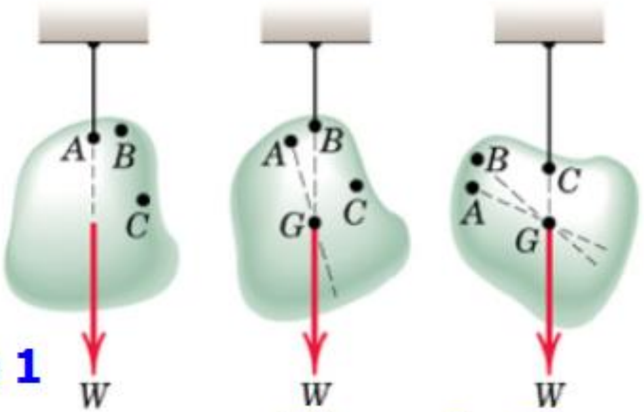


Figure 1 Figure 2 Figure 3

(1) Suspend the body from different points on the body
- Dotted lines show lines of action of the resultant force in each case.
- These lines of action will be concurrent at a single point G

As long as dimensions of the body are smaller compared with those of the earth.
- we assume uniform and parallel force field due to the gravitational attraction of the earth.

∗ The unique **Point G** is called the Center of Gravity of the body (CG)

Determination of Centre of Gravity

Determination of CG

- Apply Principle of Moments

Moment of resultant gravitational force W about any axis equals sum of the moments about the same axis of the gravitational forces dW acting on all particles treated as infinitesimal elements.

Weight of the body $W = \int dW$ **(i)**

Moment of weight of an element (dW) @ x-axis = ydW **(ii)**

Sum of moments for all elements of body = $\int ydW$ **(iii)**

From Principle of Moments: $\int ydW = \bar{y}\,W$ **(iv)**

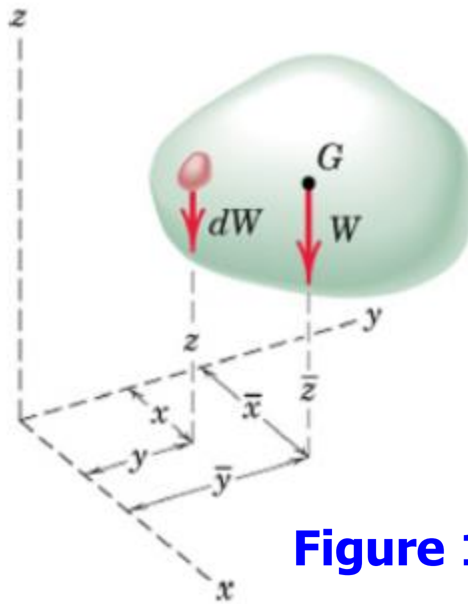


Figure 1

$$\bar{x} = \frac{\int xdW}{W} \quad \bar{y} = \frac{\int ydW}{W} \quad \bar{z} = \frac{\int zdW}{W}$$ **(v)**

→ Numerator of these expressions represents the sum of the moments; Product of W and corresponding coordinate of G represents the moment of the sum → Moment Principle.

Determination of Centre of Gravity *(contd.)*

Determination of CG

Substituting $W = mg$ and $dW = gdm$

→ $$\bar{x} = \frac{\int x\,dm}{m} \quad \bar{y} = \frac{\int y\,dm}{m} \quad \bar{z} = \frac{\int z\,dm}{m}$$ **(i)**

In vector notations:

Position vector for elemental mass: $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ **(ii)**

Position vector for mass center G: $\bar{\mathbf{r}} = \bar{x}\mathbf{i} + \bar{y}\mathbf{j} + \bar{z}\mathbf{k}$ **(iii)**

→ $$\bar{\mathbf{r}} = \frac{\int \mathbf{r}\,dm}{m}$$ **(iv)**

The above equations are the components of this single vector equation

Density ρ of a body = **mass per unit volume**

→ Mass of a differential element of volume dV → $dm = \rho dV$ **(v)**

→ ρ may not be constant throughout the body

$$\bar{x} = \frac{\int x\rho\,dV}{\int \rho\,dV} \quad \bar{y} = \frac{\int y\rho\,dV}{\int \rho\,dV} \quad \bar{z} = \frac{\int z\rho\,dV}{\int \rho\,dV}$$ **(vi)**

$$\bar{x} = \frac{\int x\,dW}{W} \quad \bar{y} = \frac{\int y\,dW}{W} \quad \bar{z} = \frac{\int z\,dW}{W}$$

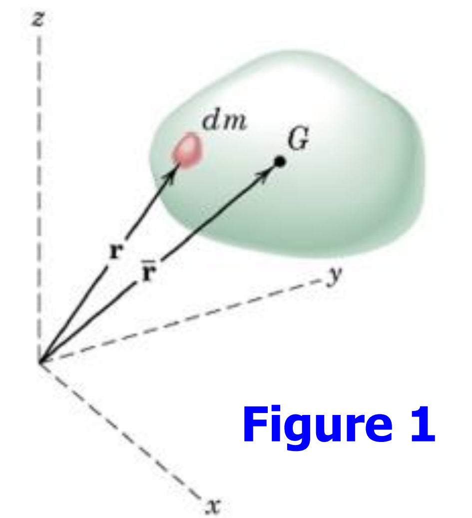


Figure 1

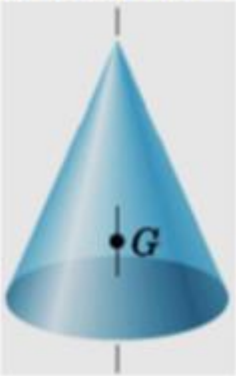
Centre of Mass

Center of Mass: Following equations independent of g

$$\bar{x} = \frac{\int x\,dm}{m} \quad \bar{y} = \frac{\int y\,dm}{m} \quad \bar{z} = \frac{\int z\,dm}{m} \qquad \bar{\mathbf{r}} = \frac{\int \mathbf{r}\,dm}{m} \qquad \bar{x} = \frac{\int x\rho\,dV}{\int \rho\,dV} \quad \bar{y} = \frac{\int y\rho\,dV}{\int \rho\,dV} \quad \bar{z} = \frac{\int z\rho\,dV}{\int \rho\,dV} \qquad \text{(i)}$$

→ They define a unique point, which is a function of distribution of mass
→ This point is Center of Mass (CM)
→ CM coincides with CG as long as gravity field is treated as uniform and parallel
→ CG or CM may lie outside the body

CM always lie on a line or a plane of symmetry in a homogeneous body

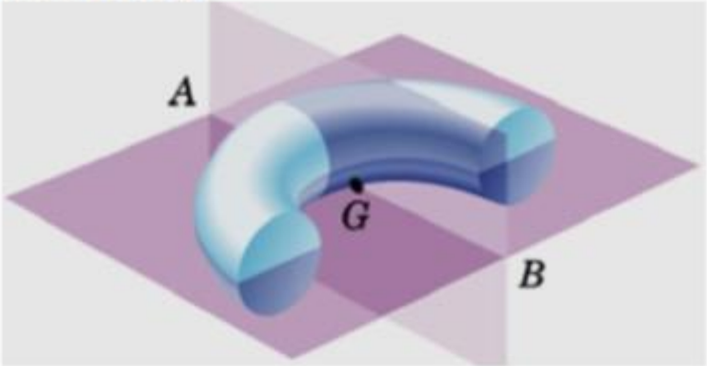


Right Circular Cone
CM on central axis

Figure 1

Half Right Circular Cone
CM on vertical plane of symmetry

Figure 2



Half Ring
CM on intersection of two planes of symmetry (line AB)

Figure 3

Centroids of Lines, Areas & Volumes

$$\bar{x} = \frac{\int x\rho dV}{\int \rho dV} \quad \bar{y} = \frac{\int y\rho dV}{\int \rho dV} \quad \bar{z} = \frac{\int z\rho dV}{\int \rho dV} \qquad \text{(i)}$$

Centroid is a geometrical property of a body

→ When density of a body is uniform throughout, centroid and CM coincide

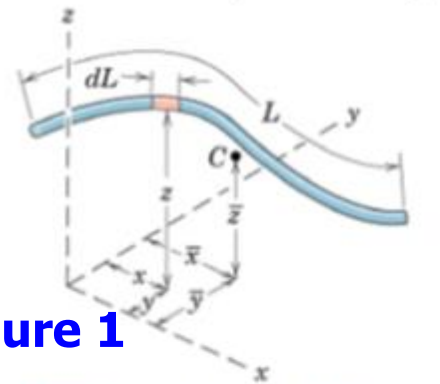


Figure 1

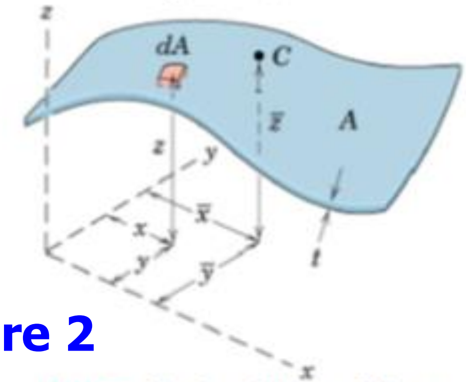


Figure 2

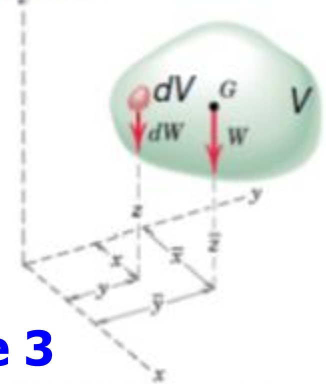


Figure 3

Lines: Slender rod, Wire
Cross-sectional area = A
ρ and A are constant over L
$dm = \rho A dL$; Centroid = CM

$$\bar{x} = \frac{\int x dL}{L} \quad \bar{y} = \frac{\int y dL}{L} \quad \bar{z} = \frac{\int z dL}{L} \qquad \text{(ii)}$$

Areas: Body with small but constant thickness t
Cross-sectional area = A
ρ and A are constant over A
$dm = \rho t dA$; Centroid = CM

$$\bar{x} = \frac{\int x dA}{A} \quad \bar{y} = \frac{\int y dA}{A} \quad \bar{z} = \frac{\int z dA}{A} \qquad \text{(iii)}$$

Numerator = First moments of Area

Volumes: Body with volume V
ρ constant over V
$dm = \rho dV$ Centroid = CM

$$\bar{x} = \frac{\int x dV}{V} \quad \bar{y} = \frac{\int y dV}{V} \quad \bar{z} = \frac{\int z dV}{V} \qquad \text{(iv)}$$

Determination of Centroids by Integration

$$\bar{x}A = \int x\,dA = \iint x\,dx\,dy = \int \bar{x}_{el}\,dA \quad \textbf{(i)}$$

$$\bar{y}A = \int y\,dA = \iint y\,dx\,dy = \int \bar{y}_{el}\,dA \quad \textbf{(ii)}$$

- Double integration to find the first moment may be **avoided by defining *dA* as a thin rectangle or strip**.

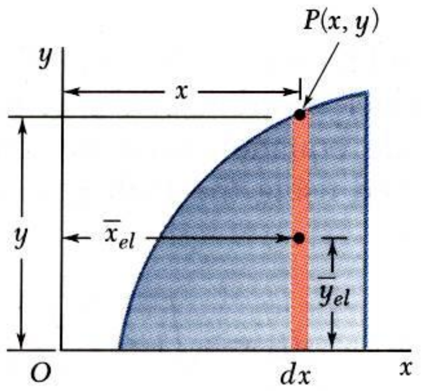


Figure 1

$$\bar{x}A = \int \bar{x}_{el}\,dA \quad \textbf{(i)}$$
$$= \int x\left(y\,dx\right)$$
$$\bar{y}A = \int \bar{y}_{el}\,dA \quad \textbf{(ii)}$$
$$= \int \frac{y}{2}\left(y\,dx\right)$$

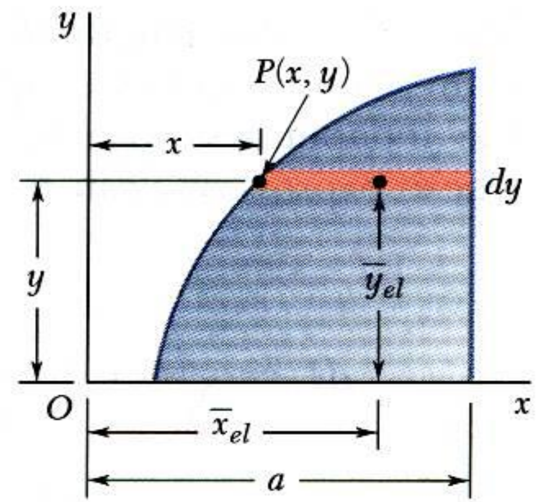


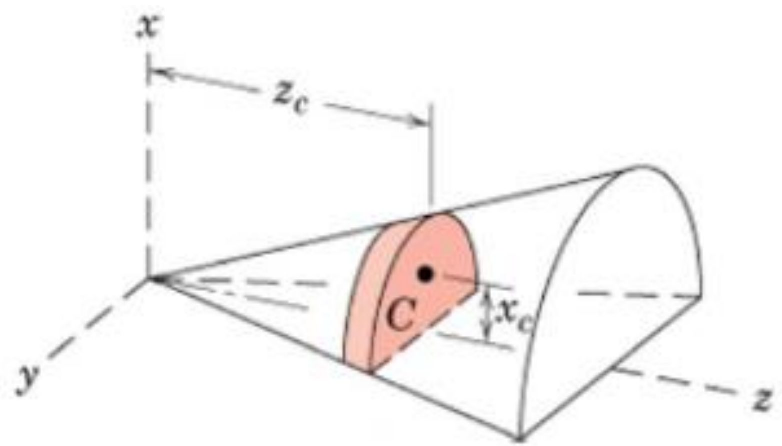
Figure 2

$$\bar{x}A = \int \bar{x}_{el}\,dA \quad \textbf{(iii)}$$
$$= \int \frac{a+x}{2}\left[(a-x)dy\right] \quad \textbf{(iv)}$$
$$\bar{y}A = \int \bar{y}_{el}\,dA$$
$$= \int y\left[(a-x)dy\right]$$

Guideline for Determination of Centroids by Integration

- Centroidal Coordinate of Differential Elements

  While expressing moment of differential elements, take coordinates of the centroid of the differential element as lever arm
  (not the coordinate describing the boundary of the area)



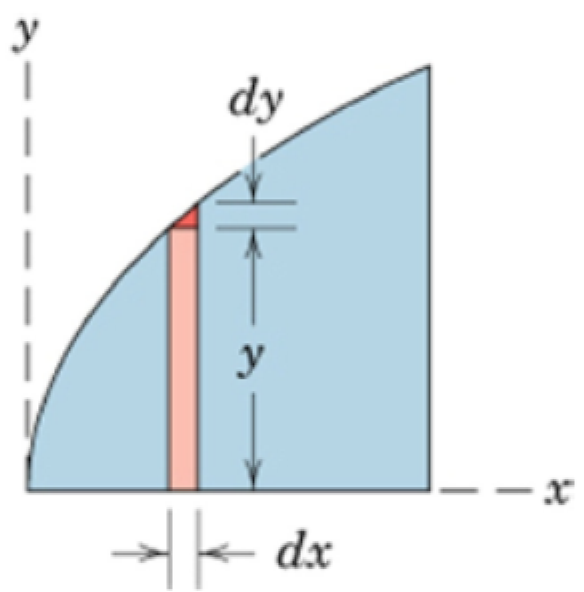
$$\bar{x} = \frac{\int x_c dV}{V} \quad \bar{y} = \frac{\int y_c dV}{V} \quad \bar{z} = \frac{\int z_c dV}{V} \quad \textbf{(i)}$$

Guidelines for choice of Integration of different regions (*contd.*)

- Discarding Higher Order Terms

  Higher order terms may always be dropped compared with lower order terms

Vertical strip of area under the curve is given by the first order term → $dA = ydx$
The second order triangular area $0.5dxdy$ may be discarded



Center of Mass and Centroids

Examples: Centroids

Locate the centroid of the circular arc

Solution: Polar coordinate system is better
Since the figure is symmetric: centroid lies on the x axis

Differential element of arc has length $dL = rd\theta$ **(a)**
Total length of arc: $L = 2\alpha r$ **(b)**
x-coordinate of the centroid of differential element: $x = r\cos\theta$

$$\bar{x} = \frac{\int x dL}{L} \quad \bar{y} = \frac{\int y dL}{L} \quad \bar{z} = \frac{\int z dL}{L}$$ **(i)**

$$L\bar{x} = \int x dL \qquad 2\alpha r\bar{x} = \int_{-\alpha}^{\alpha} (r\cos\theta) r d\theta$$ **(ii)**

$$2\alpha r\bar{x} = 2r^2 \sin\alpha$$ **(iii)**

$$\bar{x} = \frac{r\sin\alpha}{\alpha}$$ **(iv)**

For a semi-circular arc: $2\alpha = \pi$ → centroid lies at $2r/\pi$

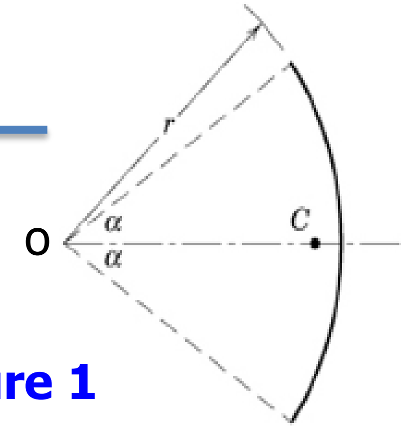


Figure 1

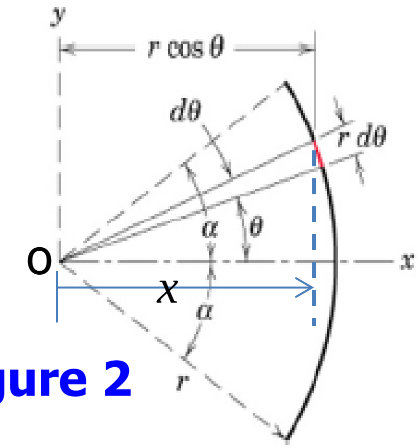


Figure 2

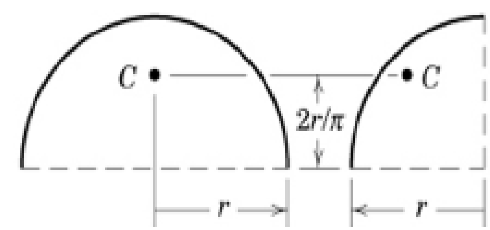
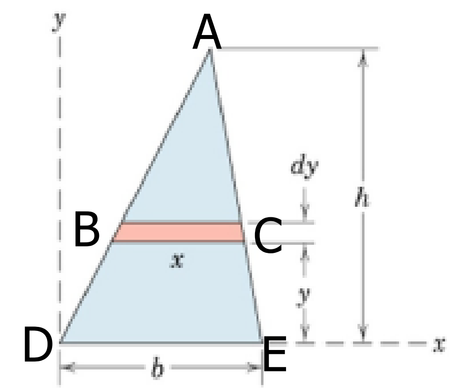


Figure 3

Center of Mass and Centroids

Examples: Centroids

Locate the centroid of the triangle along h from the base



Solution:

$$dA = x\,dy \qquad \frac{x}{(h-y)} = \frac{b}{h} \qquad \textbf{(i)}$$

Total Area $A = \frac{1}{2}bh \qquad y = y_c$ **(ii)**

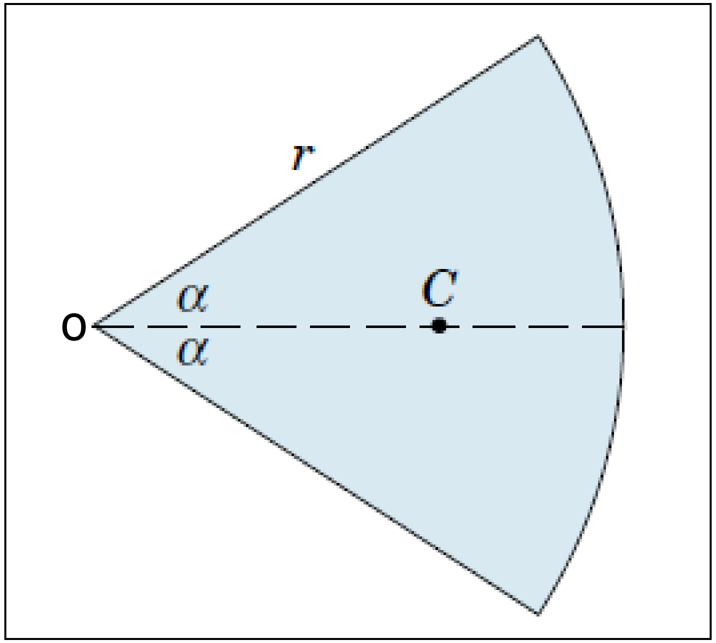
$$\bar{x} = \frac{\int x_c\,dA}{A} \quad \bar{y} = \frac{\int y_c\,dA}{A} \quad \bar{z} = \frac{\int z_c\,dA}{A} \qquad \textbf{(iii)}$$

$$A\bar{y} = \int y_c\,dA \quad \Rightarrow \frac{bh}{2}\bar{y} = \int_0^h y\frac{b(h-y)}{y}dy = \frac{bh^2}{6} \qquad \textbf{(iv)}$$

$$\boxed{\bar{y} = \frac{h}{3}} \qquad \textbf{(v)}$$

Problem 3

Centroid of the area of a circular sector. Locate the centroid of the area of a circular sector with respect to its vertex.



Solution of Problem 3 *(contd.)*

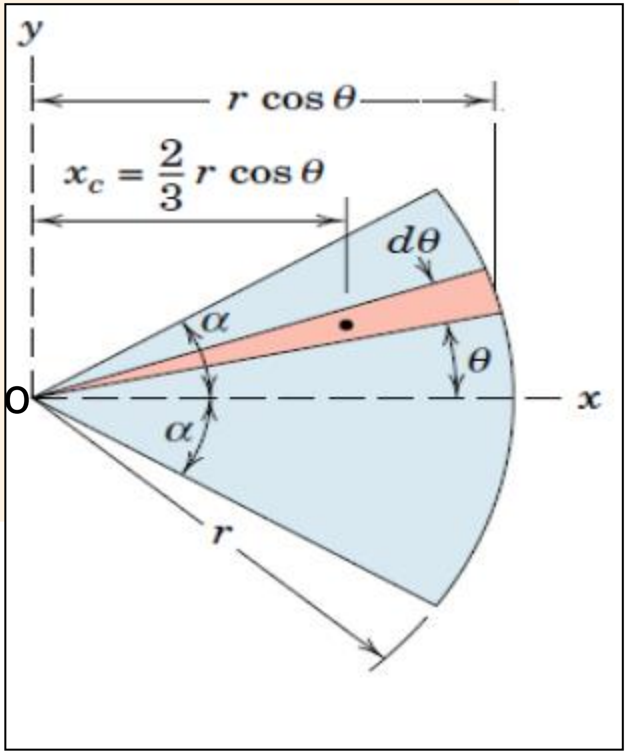
Solution The area may be covered by swinging a triangle of differential area about the vertex and through the total angle of the sector. This triangle, shown in the illustration, has an area $dA = (r/2)(r\,d\theta)$, where higher-order terms are neglected.

The centroid of the triangular element of area is two-thirds of its altitude from its vertex, so that the x-coordinate to the centroid of the element is $x_c = \frac{2}{3}r\cos\theta$. **(i)**

$$\left[A\bar{x} = \int x_c\,dA\right] \quad \textbf{(ii)} \qquad (r^2\alpha)\bar{x} = \int_{-\alpha}^{\alpha} \left(\tfrac{2}{3}r\cos\theta\right)\left(\tfrac{1}{2}r^2\,d\theta\right) \quad \textbf{(iii)}$$

$$r^2\alpha\bar{x} = \tfrac{2}{3}r^3\sin\alpha$$

$$\bar{x} = \frac{2}{3}\frac{r\sin\alpha}{\alpha} \quad \textbf{(iv)}$$



Problem 4

Hemispherical volume. Locate the centroid of the volume of a hemisphere of radius r with respect to its base.

Solution I. With the axes chosen as shown in the figure, $\bar{x} = \bar{z} = 0$ by symmetry. The most convenient element is a circular slice of thickness dy parallel to the x-z plane. Since the hemisphere intersects the y-z plane in the circle $y^2 + z^2 = r^2$, the radius of the circular slice is $z = +\sqrt{r^2 - y^2}$. The volume of the elemental slice becomes

$$dV = \pi(r^2 - y^2)\,dy \quad \textbf{(i)}$$

$$[V\bar{y} = \int y_c\,dV] \quad \textbf{(ii)} \quad \bar{y}\int_0^r \pi(r^2 - y^2)\,dy = \int_0^r y\pi(r^2 - y^2)\,dy$$

(iii)

where $y_c = y$. Integrating gives

$$\tfrac{2}{3}\pi r^3\bar{y} = \tfrac{1}{4}\pi r^4 \qquad \bar{y} = \tfrac{3}{8}r \quad \textbf{(iv)}$$

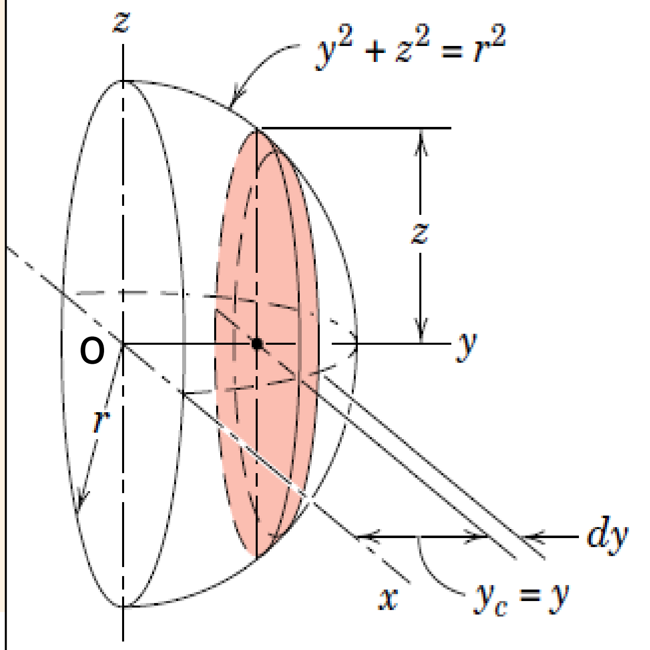
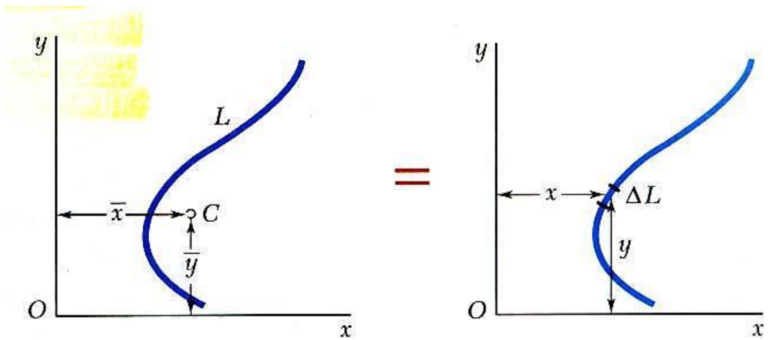


Figure 1

Centroids & First Moments of Lines

- Centroid of a line



$$\bar{x}W = \int x\,dW \qquad \textbf{(i)}$$

$$\bar{x}(\gamma La) = \int x(\gamma a)dL \qquad \textbf{(ii)}$$

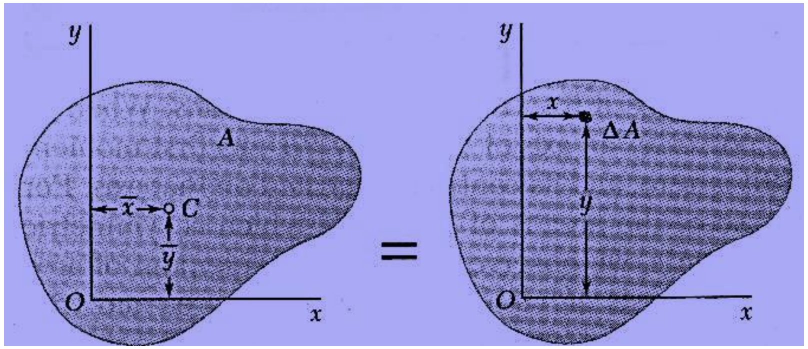
$$\bar{x}L = \int x\,dL \qquad \textbf{(iii)}$$

$$\bar{y}L = \int y\,dL \qquad \textbf{(iv)}$$

where γ = specific weight (weight per unit volume) and a = cross-sectional area

Centroids & First Moments of Areas

- Centroid of an area



$$\bar{x}W = \int x\,dW \qquad \textbf{(i)}$$

$$\bar{x}(\gamma A t) = \int x(\gamma t)\,dA \qquad \textbf{(ii)}$$

$$\bar{x}A = \int x\,dA = Q_y \qquad \textbf{(iii)}$$

$$= \text{first moment with respect to } y$$

$$\bar{y}A = \int y\,dA = Q_x \qquad \textbf{(iv)}$$

$$= \text{first moment with respect to } x$$

where γ = density or specific weight (weight per unit volume), t = thickness

First Moments of Areas and Lines

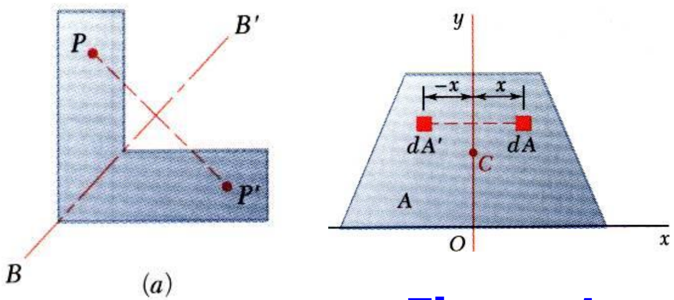


Figure 1

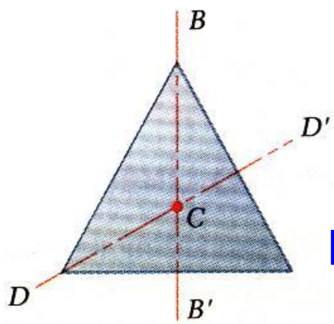


Figure 2

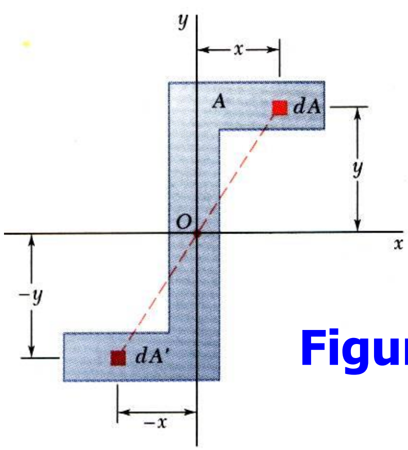
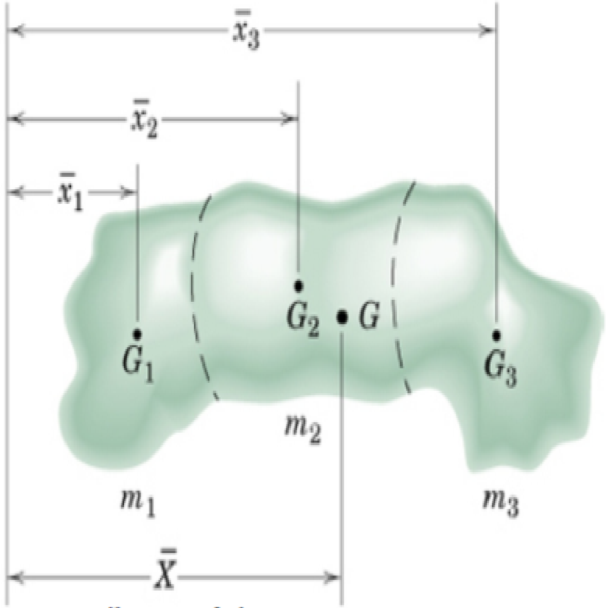


Figure 3

1. An area is symmetric with respect to an axis *BB'* if for every point *P* there exists a point *P'* such that *PP'* is perpendicular to *BB'* and is divided into two equal parts by *BB'*.
2. **The first moment of an area with respect to a line of symmetry is zero.**
3. If an area possesses a line of symmetry, its centroid lies on that axis
4. If an area possesses two lines of symmetry, its centroid lies at their intersection.
5. An area is symmetric with respect to a center *O* if for every element *dA* at (*x*,*y*) there exists an area *dA'* of equal area at (-*x*,-*y*).
6. The centroid of the area coincides with the center of symmetry.

Centre of Mass & Centroids of Composite bodies

Divide bodies or figures into several parts such that their mass centers can be conveniently determined
→ Use Principle of Moment for all finite elements of the body



x-coordinate of the center of mass of the whole

$$(m_1 + m_2 + m_3)\bar{X} = m_1\bar{x}_1 + m_2\bar{x}_2 + m_3\bar{x}_3 \quad \textbf{(i)}$$

Mass Center Coordinates can be written as:

$$\bar{X} = \frac{\sum m\bar{x}}{\sum m} \quad \bar{Y} = \frac{\sum m\bar{y}}{\sum m} \quad \bar{Z} = \frac{\sum m\bar{z}}{\sum m}$$

m's can be replaced by *L*'s, *A*'s, and *V*'s for lines, areas, and volumes

Integration vs Approx. Summation of Irregular bodies *(contd.)*

Reduce the problem to one of locating the centroid of area

→ Appx Summation may be used instead of integration

Figure 1

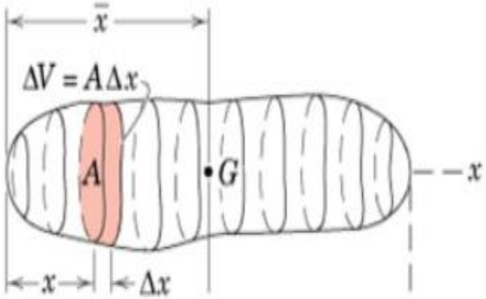
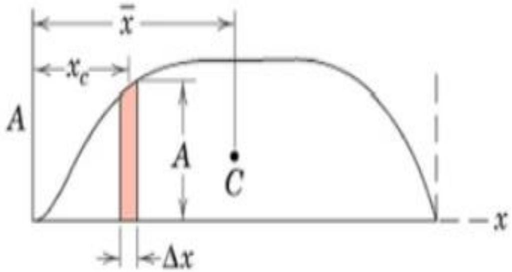


Figure 2



Divide the area into several strips

Volume of each strip = $A\Delta x$

Plot all such A against x.

→ Area under the plotted curve represents volume of whole body and the x-coordinate of the centroid of the area under the curve is given by:

$$\bar{x} = \frac{\sum (A\Delta x)x_c}{\sum A\Delta x} \Rightarrow \bar{x} = \frac{\sum V x_c}{\sum V}$$

Accuracy may be improved by reducing the width of the strip

Composite Plates and Areas

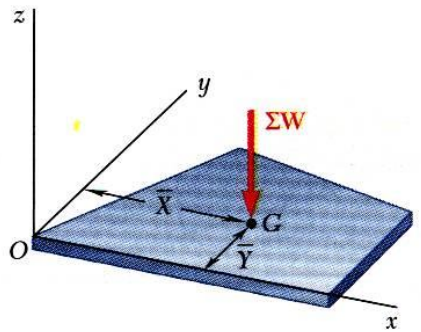


Figure 1

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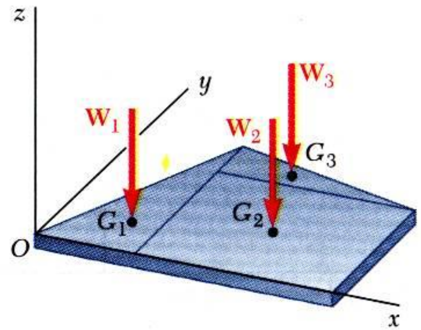

Figure 2

- Composite plates

$$\overline{X}\sum W = \sum \overline{x}W \qquad \textbf{(i)}$$

$$\overline{Y}\sum W = \sum \overline{y}W \qquad \textbf{(ii)}$$

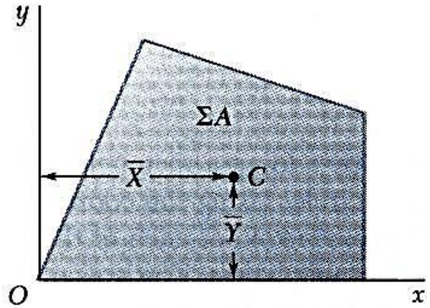

Figure 3

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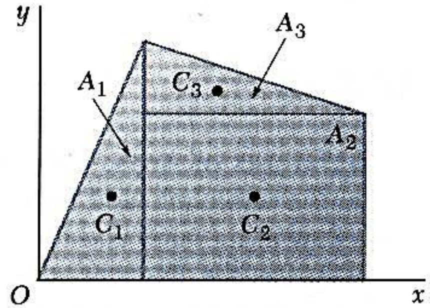


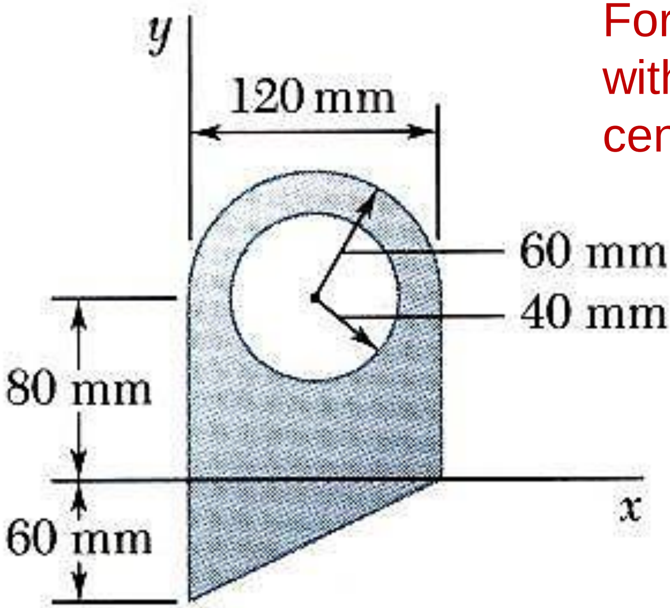
Figure 4

- Composite area

$$\overline{X}\sum A = \sum \overline{x}A \qquad \textbf{(iii)}$$

$$\overline{Y}\sum A = \sum \overline{y}A \qquad \textbf{(iv)}$$

Sample Problem



For the plane area shown, determine the first moments with respect to the *x* and *y* axes and the location of the centroid.

SOLUTION:

1. Divide the area into a triangle, rectangle, and semicircle with a circular cutout.

2. Calculate the first moments of each area with respect to the axes.

3. Find the total area and first moments of the triangle, rectangle, and semicircle. Subtract the area and first moment of the circular cutout.

4. Compute the coordinates of the area centroid by dividing the first moments by the total area.

Sample Problem

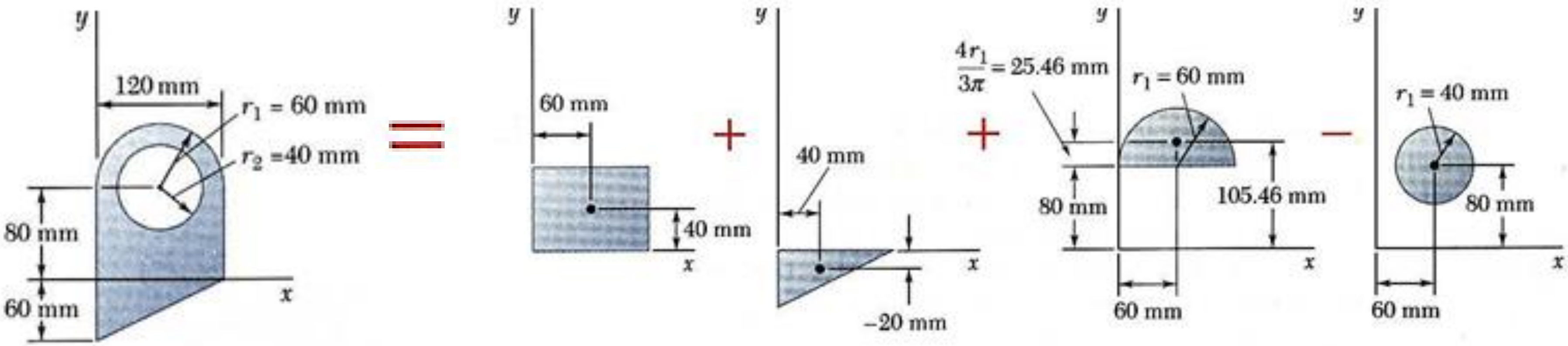


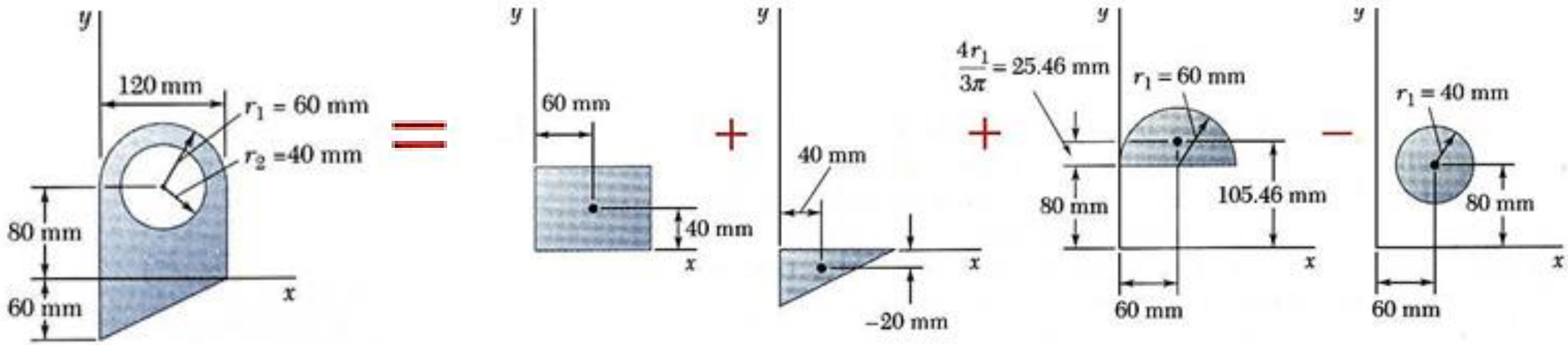
Figure 1 **Figure 2** **Figure 3** **Figure 4**

Component	A, mm²	$\bar{x}$, mm	$\bar{y}$, mm	$\bar{x}A$, mm³	$\bar{y}A$, mm³
Rectangle	$(120)(80) = 9.6 \times 10^3$	60	40	$+576 \times 10^3$	$+384 \times 10^3$
Triangle	$\frac{1}{2}(120)(60) = 3.6 \times 10^3$	40	−20	$+144 \times 10^3$	-72×10^3
Semicircle	$\frac{1}{2}\pi(60)^2 = 5.655 \times 10^3$	60	105.46	$+339.3 \times 10^3$	$+596.4 \times 10^3$
Circle	$-\pi(40)^2 = -5.027 \times 10^3$	60	80	-301.6×10^3	-402.2×10^3
	$\Sigma A = 13.828 \times 10^3$			$\Sigma\bar{x}A = +757.7 \times 10^3$	$\Sigma\bar{y}A = +506.2 \times 10^3$

- Find the **total area** and **first moments** of the triangle, rectangle, and semicircle. **Subtract the area and first moment of the circular cutout.**

$$\boxed{\begin{aligned} Q_x &= +506.2\times10^3\,\text{mm}^3 \\ Q_y &= +757.7\times10^3\,\text{mm}^3 \end{aligned}}$$

Sample Problem



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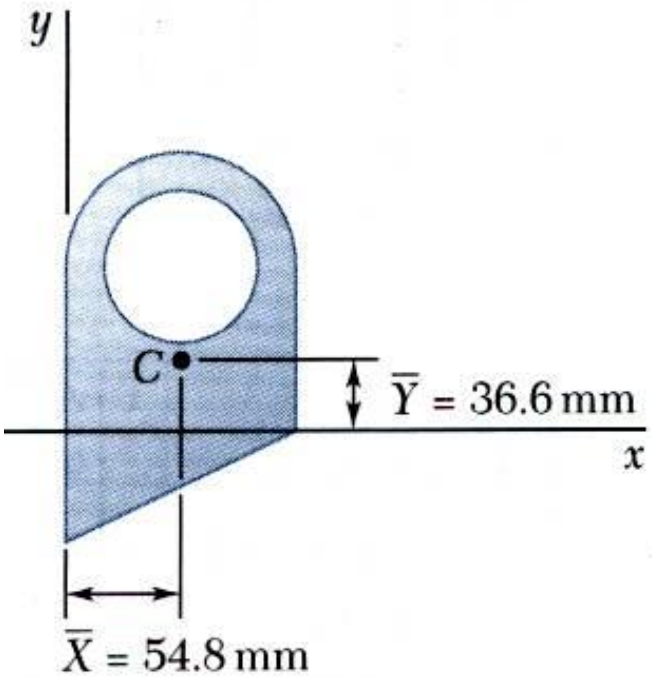
- Find the **total area** and **first moments** of the triangle, rectangle, and semicircle. **Subtract the area and first moment of the circular cutout.**

$$Q_x = +506.2 \times 10^3\,\text{mm}^3$$

$$Q_y = +757.7 \times 10^3\,\text{mm}^3$$

Sample Problem

Compute the coordinates of the area centroid by dividing the first moments by the total area.



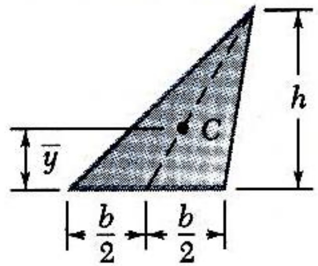
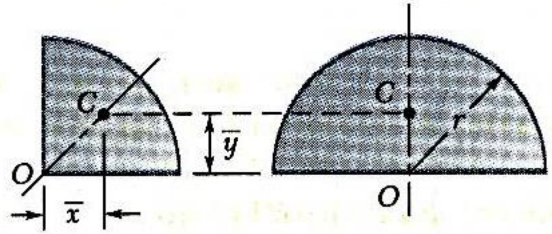
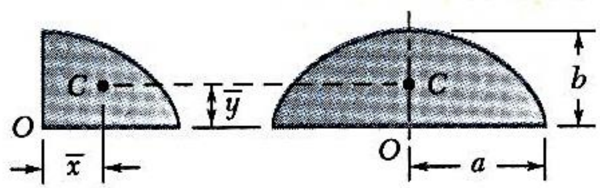
$$\bar{X} = \frac{\sum \bar{x} A}{\sum A} = \frac{+757.7 \times 10^3 \text{ mm}^3}{13.828 \times 10^3 \text{ mm}^2} \qquad \textbf{(i)}$$

$$\boxed{\bar{X} = 54.8 \text{ mm}}$$

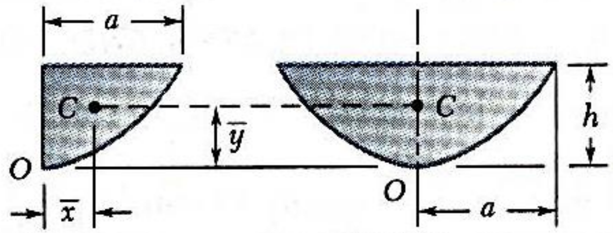
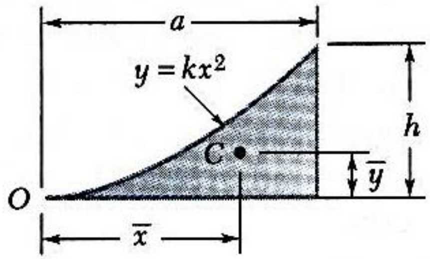
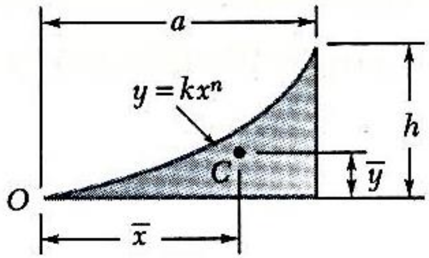
$$\bar{Y} = \frac{\sum \bar{y} A}{\sum A} = \frac{+506.2 \times 10^3 \text{ mm}^3}{13.828 \times 10^3 \text{ mm}^2} \qquad \textbf{(ii)}$$

$$\boxed{\bar{Y} = 36.6 \text{ mm}}$$

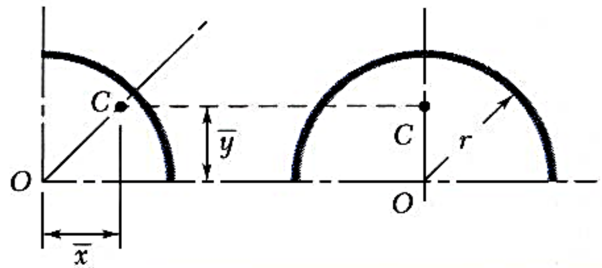
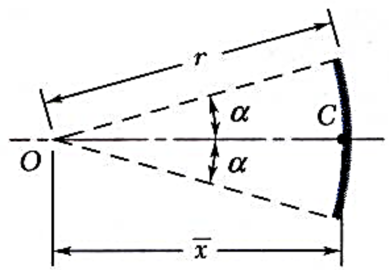
Centroids of Common Shapes of Areas

Shape		$\bar{x}$	$\bar{y}$	Area
Triangular area			$\frac{h}{3}$	$\frac{bh}{2}$
Quarter-circular area		$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{4}$
Semicircular area		0	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{2}$
Quarter-elliptical area		$\frac{4a}{3\pi}$	$\frac{4b}{3\pi}$	$\frac{\pi ab}{4}$
Semielliptical area		0	$\frac{4b}{3\pi}$	$\frac{\pi ab}{2}$

Centroids of Common Shapes of Areas

Semiparabolic area		$\frac{3a}{8}$	$\frac{3h}{5}$	$\frac{2ah}{3}$
Parabolic area		0	$\frac{3h}{5}$	$\frac{4ah}{3}$
Parabolic spandrel		$\frac{3a}{4}$	$\frac{3h}{10}$	$\frac{ah}{3}$
General spandrel		$\frac{n+1}{n+2}a$	$\frac{n+1}{4n+2}h$	$\frac{ah}{n+1}$

Centroids of Common Shapes of Lines

Shape		$\bar{x}$	$\bar{y}$	Length
Quarter-circular arc		$\frac{2r}{\pi}$	$\frac{2r}{\pi}$	$\frac{\pi r}{2}$
Semicircular arc		0	$\frac{2r}{\pi}$	πr
Arc of circle		$\frac{r \sin \alpha}{\alpha}$	0	$2\alpha r$

End